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The impact of policy collaboration networks and policy topic networks on policy diffusion: Empirical evidence from the energy field

Jiaxi Wang, Jingjing Zhang *

School of Public Policy and Management, University of Chinese Academy of Sciences, Beijing, China

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ABSTRACT

Policy networks have strong explanatory power for policy processes and policy results. Previous studies have proven the importance of the role played by policy collaboration networks. Recently, the value of policy topic networks has gradually been revealed. However, there has been a lack of interpretation of policy results through topic network characteristics. Therefore, a comparative research framework for policy collaboration and topic networks is proposed. Based on a search and screening results for China's energy policies, we construct a policy collaboration network composed of policymaking agencies and their collaborations and a topic network composed of policy topics and their co-occurrences. After analysing the characteristics of institutions and topics, we empirically test hypotheses concerning the relationship between the network characteristics of these two networks and policy diffusion in China. We find that in policy collaboration networks, the centrality of agencies has a positive effect on the degree of policy diffusion and the structural holes of agencies exert an inverted U-shaped effect on the degree of policy diffusion. In the topic networks, policy topics with high centrality have a greater degree of policy diffusion and the structural holes of topics are not significantly associated with policy diffusion.

1. Introduction

Policy networks have long been a focus of research. A policy network is the relationship structure formed by interest correlations among social participants in the policy process (Marsh and Rhodes, 1992). Policy network theory is usually used to analyse the interaction among policy participants and the impact of this interaction on policy outcomes (Yu et al., 2022). The characteristics of policy networks have been proven to play an important role in the formulation, implementation and consequences of policies (Wagner et al., 2021; Yi, 2018; Yu et al., 2022). Building on this literature, recent studies have begun to focus on the importance of another kind of network: the policy topic network (Chen et al., 2022; Wang and Zhang, 2022; Yang and Huang, 2020). This network theory, which is based on topic co-occurrence, reveals the policy focus and the associations of policy elements in the field, providing more comprehensive insight into policy content (Ba et al., 2023). Energy is the source of power for human production and life, and is the foundation of social and economic development (Kong et al., 2020; Zeng et al., 2021). With the vigorous development of the energy industry, it is important to conduct a dual analysis of the consequent energy policy and topic networks. The results can assist policymakers in grasping the characteristics of energy policy evolution, enhancing their understanding of energy development and providing a reference for effective decision-making.

Policy diffusion is a dynamically evolving process of policy learning in which adopters absorb policy content and achieve policy innovation by imitating, reinventing, or reusing existing policies (Wang et al., 2020). Most studies use the adoption of policies as a measure of policy diffusion (Hrovatin et al., 2021; Raghoo and Shah, 2022). However, this approach ignores the time effects inherent in policy evolution (Wu and Han, 2022). In recent years, some scholars have attempted to refine the meaning of policy diffusion by supplementing it with the diffusion rate or extent, thus studying the dynamic innovation process of policy diffusion from the perspective of policy evolution (Ba et al., 2022; Hu et al., 2020). The need for collaboration arises when policy learning does not meet the needs of the target (Wang et al., 2020). Collaboration among policy actors is an important driver of policy innovation, and resource sharing facilitates more effective pursuit of policy goals (Liu and Xu, 2018). Therefore, it is reasonable to combine policy diffusion with collaboration theory.

* Corresponding author.
E-mail addresses: wangjiaxi20@mails.ucas.ac.cn (J. Wang), zhangjingjing@ucas.ac.cn (J. Zhang).

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Policy networks are concerned with the interaction among interdependent actors and the impact of this interaction on organizational goals and strategies (Yu et al., 2022). Simply put, a policy network is a structured presentation of the interaction between governments. Policies are formulated through interactions among government agencies, which pursue their interests through collaboration and resource exchange (Sun and Cao, 2018). Different government agencies hold heterogeneous information resources, and policy documents serve as carriers of written and standardized resources (Wu and Han, 2022). Therefore, policy networks can reflect not only the collaborative relationship among policymaking agencies but also the exchange of information resources among those agencies. The current focus on policy networks is mainly at the level of policymaking agencies and policymakers. Scholars have analysed the impact of network attributes on policymaker behavior and organizational goals by constructing policy collaboration networks (Lundmark et al., 2021; Wang et al., 2021). On this basis, we introduce topic modeling techniques to construct policy topic networks (Wang and Zhang, 2022). Topic networks differ from collaboration networks in that policy collaboration networks use policymaking agencies as nodes and collaborative relationships among those agencies as ties. Topic networks use topics as nodes and co-occurrence relationships among those topics as ties, representing the connections among policy resources (Li et al., 2021). Linkages or breaks of these resources may have an impact on policy diffusion. We argue that to fully understand the antecedents of policy diffusion, two networks, the policy collaboration network and the policy topic network, need to be analysed.

Policy documents reflect at least two networks that are related to policy diffusion. One is the policy collaboration network, and the other is the policy topic network. Collaboration networks, as a typical social network, are widely used by policy researchers to analyse the influence of network attributes on policymakers' behavior and organizational goals (Lundmark et al., 2021; Howe et al., 2021). Klijn and Koppenjan (2000) combined network theory and network management concepts to propose a policy network governance approach. This approach confirmed the important explanatory power of the policy network framework at the agency relationship level. Yi (2018) investigated the relationship between policy collaboration network structure and governance performance and showed that nodes in the network with high centrality and clustering coefficients performed better than other nodes in the network. Yu et al., 2022) revealed the complex network structure formed by policy actors and their interactions and discussed the influence of network characteristics on the policymaking process. Therefore, policy collaboration networks are important for explaining the relationship among government agencies and the resulting impact on policy outcomes. However, a policy involves not only collaboration networks but also topic networks.

Policy topic networks are formed at the level of policy texts. High-frequency words that appear in policy documents (i.e., corpora) can reflect the central message of the policy; these high-frequency words are referred to as "topics" (Richarz et al., 2023; Yang and Huang, 2020). This approach has become a research tool for many scholars (Wallmeier et al., 2019; Gülzau, 2020). Yang and Huang (2020) identified and extracted topics of policy objectives from policy documents and revealed dynamic changes in policies through the establishment of a topic co-occurrence network. Feng et al. (2020) combined the networks to estimate the potential relationships and topics associated with electric vehicles and found that battery protection was a very promising topic. Furthermore, Gan et al. (2023) analysed dynamic changes in China's green building policies based on a policy network, revealing the continuity of changes in policy objectives and the imbalance in the adoption of policy tools. Through a review of the relevant literature, we find that most research on policy topic networks has aimed to reveal the law of policy evolution. However, few studies have explored the impact of policy topic networks on policy diffusion. This gap inspired our study of topic networks.

Researchers are increasingly using policy collaboration networks or topic networks to analyse the pattern characteristics and evolutionary trajectory of energy development. A collaboration network focuses on the collaboration of core agencies and their impact on energy development. Tang (2018) found that collaboration among wind power participants is conducive to improving the performance of wind farms. Wagner et al. (2021) demonstrated that government agencies are the most influential actors in these networks, and their central position in the collaboration network further enhances their influence. A topic network focuses on revealing important energy topics and developments. By identifying the topic and reference relationships of China's new energy policy, Ba et al. (2023) revealed the dynamic trajectory of policy communication. Richarz et al. (2023) explored the evolution of the German energy topic network and found that wind and solar dominated. Furthermore, Wang and Zhang (2022) found that the structural characteristics and edge attributes of a policy topic network had a positive impact on policy recombinant creation. However, the application of the policy network method in the energy field is relatively new. Studies examining the effects of policy collaboration and topic networks on policy performance are rare. We hope to fill this gap by investigating the different impacts of these two networks on policy diffusion.

Through a review of policy collaboration and topic networks, we find that most recent policy network studies have focused on agencies or policymakers (Lundmark et al., 2021; Wang et al., 2021). The framework for using policy collaboration networks to research participant characteristics and interactive relationships is reaching maturity. However, comparatively less attention has been given to the content and structure of policy texts. Although recent studies have recognized the importance of policy topics, they have tended to focus on exploring the evolution of policy topics and identifying policy core issues while ignoring the significance of topic networks in explaining policy outcomes. In addition, we found a lack of related discussion on energy policy collaboration and topic networks in the energy field, which could lead to a fragmentation of energy policy analysis. To address this issue, we conducted a comparative analysis of the impact of collaboration and topic networks on policy diffusion in the context of Chinese energy politics.

This study constructs a policy collaboration network that uses government agencies as nodes and a policy topic network that uses policy topics as nodes. As policy documents are updated over time, the relationship among policymaking agencies changes accordingly, as is the case for policy topics. This change process leads to the formation and development of policy collaboration networks and policy topic networks. We propose that the node attributes of these two networks exert an impact on policy diffusion. Importantly, these two networks are decoupled. In other words, the position of policymakers in the collaboration network may not match the position of policy topics in the topic network (Wang et al., 2014). The degree of prestige, as represented by centrality, and the advantage of nonredundant information control, as represented by structural holes, have both been accepted and recognized by scholars and widely cited in previous studies. Therefore, we focus on these two indicators: centrality and structural holes. We theoretically and empirically explore how centrality and structural holes within the two networks affect policy diffusion.

This paper makes three major contributions. (1) It advances the related research on the consequences of policy networks. In particular, we consider policy networks as antecedents of policy diffusion, which opens a new avenue of inquiry that offers considerable promise in explaining policy diffusion. (2) It contributes to the comprehension of the multiple natures of policy networks. By constructing policy collaboration networks consisting of government agencies and policy topic networks consisting of policy topics, we leverage a comparative perspective to explore the different impacts of the structural characteristics of these two types of networks. These networks are not isomorphic. We show that their structural characteristics are distinct

and that they influence policy diffusion in different ways through separate mechanisms. (3) By observing the evolution of core energy topics and policymaking agencies, we provide policy advice to the rapidly developing energy field in China. As the world's largest energy consumer and producer, China is a typical example of country-level energy management. The results also provide a reference for energy development in other countries.

2. Theoretical background and hypotheses

2.1. Policy collaboration network

If a node occupies a central position in a collaboration network, it will have more connections and easier access to the required information resources (Ahuja, 2000; Yi, 2018; Guan et al., 2015). These resources motivate policymaking agencies to keep up with policy innovations by providing external information and new ideas. At the same time, the concept of centrality is associated with "power" in the field of sociology (Ingold and Leifeld, 2016). The higher the node centrality of an agency is, the greater its power, and the more the agency's authority will attract policy citations. Thus, the more central the policymaking agencies are in the collaboration network, the higher the degree of policy diffusion and the more citations those documents are likely to receive.

Hypothesis 1. The centrality in a collaboration network of the policymaking agencies of a policy document is positively related to the document's citation count.

In a policy collaboration network, nodes represent government agencies. We consider that a government agency spans a structural hole if it establishes links with other agencies that are not connected (Ahuja, 2000; Burt and Lazega, 1995). Agencies that cross structural holes have more nonredundant relationships and access to rich, heterogeneous information (Guan et al., 2017), thus improving their decision-making quality. At the same time, government agencies that occupy rich structural holes have a greater advantage in controlling the information transfer between unconnected agencies. Therefore, government agencies may obtain more citations as brokers (Rowley et al., 2000). However, Shrestha (2018) suggests that too much information sharing can reduce the relative value of information. At this point, the value of acquiring information through the pursuit of bridging links decreases, and policymaking agencies tend to pursue combinations of existing policy elements and have less time and energy to explore the leading edge of new policy creation. Government agencies may fall into a "capacity trap" (Levinthal, 1992; Zhang and Luo, 2020). The decline in the quality of policymaking leads to a decrease in the citation count.

Hypothesis 2. The structural holes in a collaboration network of the policymaking agencies of a policy document exert an inverted U-shaped effect on the document's citation count.

2.2. Decoupling policy collaboration networks from topic networks

The position of an actor in a collaboration network and the position of its knowledge in a knowledge network are not completely matched (Wang et al., 2014; Yan et al., 2022). Similarly, in the policy field, we argue that the position of a policymaking agency in a collaboration network does not necessarily match the position of its topic in the topic network. We consider the conditions under which the two networks occur in an overlapping case in which a policymaking agency corresponds to only one topic and the collaborative role of the policymaking agency involves combining the corresponding topics. However, these conditions are often not possible, as most policies contain more than one topic, topics are mostly contained in more than one policy, and collaborations between two policymaking agencies often correspond to the co-occurrence of multiple topics. Therefore, the decoupling of collaboration and topic networks is inevitable rather than accidental.

Fig. 1 visually illustrates this decoupling relationship. We assume that Policy document 1 has two policymaking agencies, a and b, and two topics, 1 and 2, and Policy document 2 has two policymaking agencies, c and d, and three topics, 3, 4 and 5. We represent this decoupling relationship by calculating the characteristic values of the two policies in the two networks. Table 1 provides the values of the centrality and structural holes in the collaboration network of Policy document 1 and Policy document 2, and Table 2 provides the values of the centrality and structural holes in the topic network of Policy document 1 and Policy document 2. Based on these two tables, we calculated the average values of the centrality and structural holes of Policy documents 1 and 2 in both networks (Table 3). The relative positions of the two policies in the two networks are different. Policy document 1 is more centralized in the collaboration network and spans more structural holes than Policy document 2. In contrast, in the topic network, Policy document 1 is less focused on topics and has more network constraints than Policy document 2. This further demonstrates that the collaboration network is decoupled from the topic network.

2.3. Policy topic network

Centrally located nodes can gain a significant communication advantage over other nodes in the network and can contact other nodes more quickly and easily (Ahuja, 2000). Thus, a centrally located topic is better positioned to create new combinations. Policymakers are more likely to seek new combination opportunities through such a topic, thus increasing policy citations. At the same time, a topic that is centrally located in a policy topic network has higher research potential and

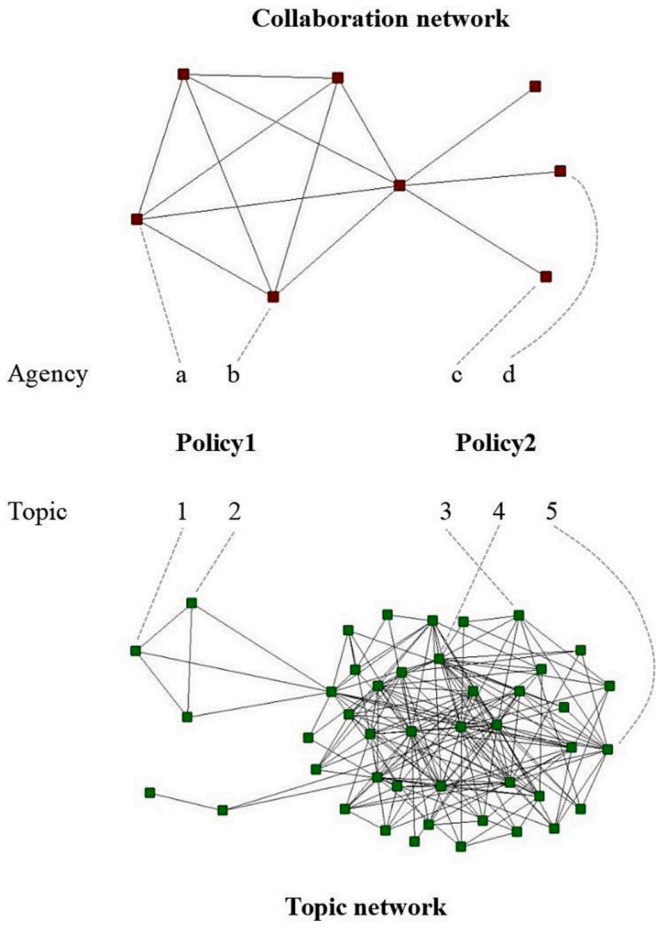


Fig. 1. Decoupling diagram between a collaboration network and a topic network.

Table 1
Description of the collaboration network indicators.

		CNNrmDegree	CNStructural holes
Policy document 1	a	0.571	1.234
	b	0.571	1.234
Policy document 2	c	0.286	0.889
	d	0.286	0.889

Table 2
Description of the topic network indicators.

		TNNrmDegree	TNStructural holes
Policy document 1	1	0.070	1.074
	2	0.070	1.074
Policy document 2	3	0.163	1.524
	4	0.442	1.839
	5	0.279	1.706

enhances policymakers' beliefs about its future development (Fleming and Sorenson, 2001; Wang et al., 2014). Importantly, we do not propose our inverted U-shaped hypothesis regarding collaboration network structural holes within a topic network. The reason is that, compared with policymaking agencies, there is no subjective awareness of the topic and the thesaurus is endless, which means that "ability trap" and "distraction" phenomena will not occur (Zhang and Chen, 2022; Yang and Huang, 2020). Therefore, policies occupying rich structural holes in a topic network positively impact the citation count.

Hypothesis 3. The average centrality in the topic network of the topics of a policy document is positively related to the document's citation count.

In a policy topic network, nodes denote topics and ties indicate the co-occurrence relationships between topics in the same policy document. If a node combines two nodes that have never previously been combined, then it is thought to span a structural hole (Burt and Lazega, 1995). The structural holes in the overall network indicate that its internal nodes also have the potential to be combined. A pair of nodes that are combined with the same mediator has more combinatorial potential than a completely unrelated pair (Wang et al., 2014). Nodes that occupy more structural holes have greater potential because they have a control advantage over nonredundant information (Burt and Lazega, 1995). In a policy topic network, if a topic spans a rich set of structural holes, then policymakers are more likely to look for opportunities for combinations among separate policy topics, thereby increasing citations of documents related to that policy. Thus, policies that occupy rich structural holes in their policy topic networks may receive more citations.

Hypothesis 4. Structural holes in the topic network of a policy document are positively related to the document's citation count.

Based on the hypotheses presented in the previous section, we have constructed the following model (Fig. 2.). We constructed the policy collaboration network and the policy topic network to capture policy-level collaborative activities. The dependent variable policy diffusion is quantified as the number of policy citations. Based on the measurement of normalized centrality and structural holes in the two networks, we propose four hypotheses: H1, H2, H3, and H4.

3. Methodology

This study focuses on the structural characteristics of policy collaboration networks and topic networks and their effects on policy diffusion. We first collected China's official energy policy documents. We extracted the metadata of these documents (e.g., policymaking agencies and citations) and extracted policy topics from the text content. Second, we constructed a policy collaboration network (i.e., a network based on the collaborative relationship of policymaking agencies) and a policy topic network (i.e., a network based on the co-occurrence relationship of policy topics). Then, we explored the evolution of policy topics and policymaking agencies. Finally, we empirically examined the influence of the structural characteristics of the collaboration network and topic network on policy diffusion. Fig. 3 summarizes our research steps.

Table 3
Comparison of mean values of network indicators in Policy document 1 and Policy document 2.

	CNNrmDegree	CNStructural holes	TNNrmDegree	TNStructural holes
Policy document 1	0.571	1.234	0.070	1.074
Policy document 2	0.286	0.889	0.295	1.690

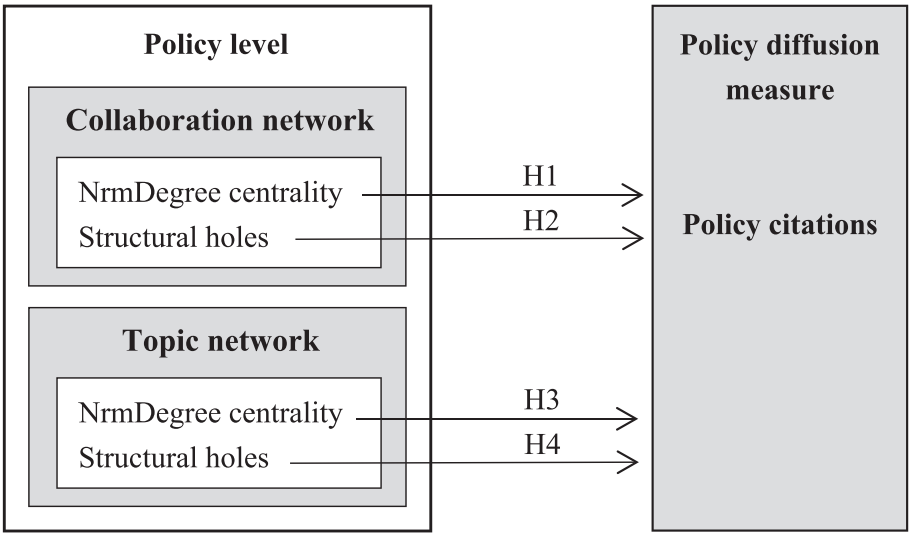


Fig. 2. Research framework.

3.1. Research context

In China, government agencies play a leading role in most policymaking processes (Sun and Cao, 2018). At the same time, Chinese society is a typical "relational society," and the policymaking process is characterized by an extremely complex network of policy relationships. Therefore, it is crucial to explore the application of the policy network method in China's policy research. We perform an empirical analysis of the energy field. Energy is the source of power for human production and life and a fundamental requirement for social and economic development (Zeng et al., 2021). With the increasing importance of the energy field and the expansion of the energy market, energy problems cannot be ignored. In China, the government tends to use policy regulation to solve certain problems in the energy field (Ma et al., 2010). Thus, the energy field is a typical example of a fast-growing and easily observable policy network.

3.2. Data collection and processing

A change in national leadership implies the advent of new leadership, which results in substantial policy changes. Therefore, we selected a period of ten years, 2008 to 2017, for our study based on the leadership change announced in the first session of the 11th National People's Congress. We screened energy-related policy documents published on official government websites during this 10-year period to create a corpus comprising 2092 text documents. The research data used in this study are taken mainly from the Laws and Regulations Database of PKULaw (https://www.pkulaw.com/), which is the most comprehensive and professional law and regulations retrieval database in China. PKULaw includes laws, administrative regulations, judicial interpretations, departmental rules, group regulations and industry regulations. First, we searched for energy-related policy documents from 2008 to 2017 in PKULaw using the title search term "energy," thus creating a dataset containing 2092 policy documents. Notably, a policy document obtained from this website contains both its text and

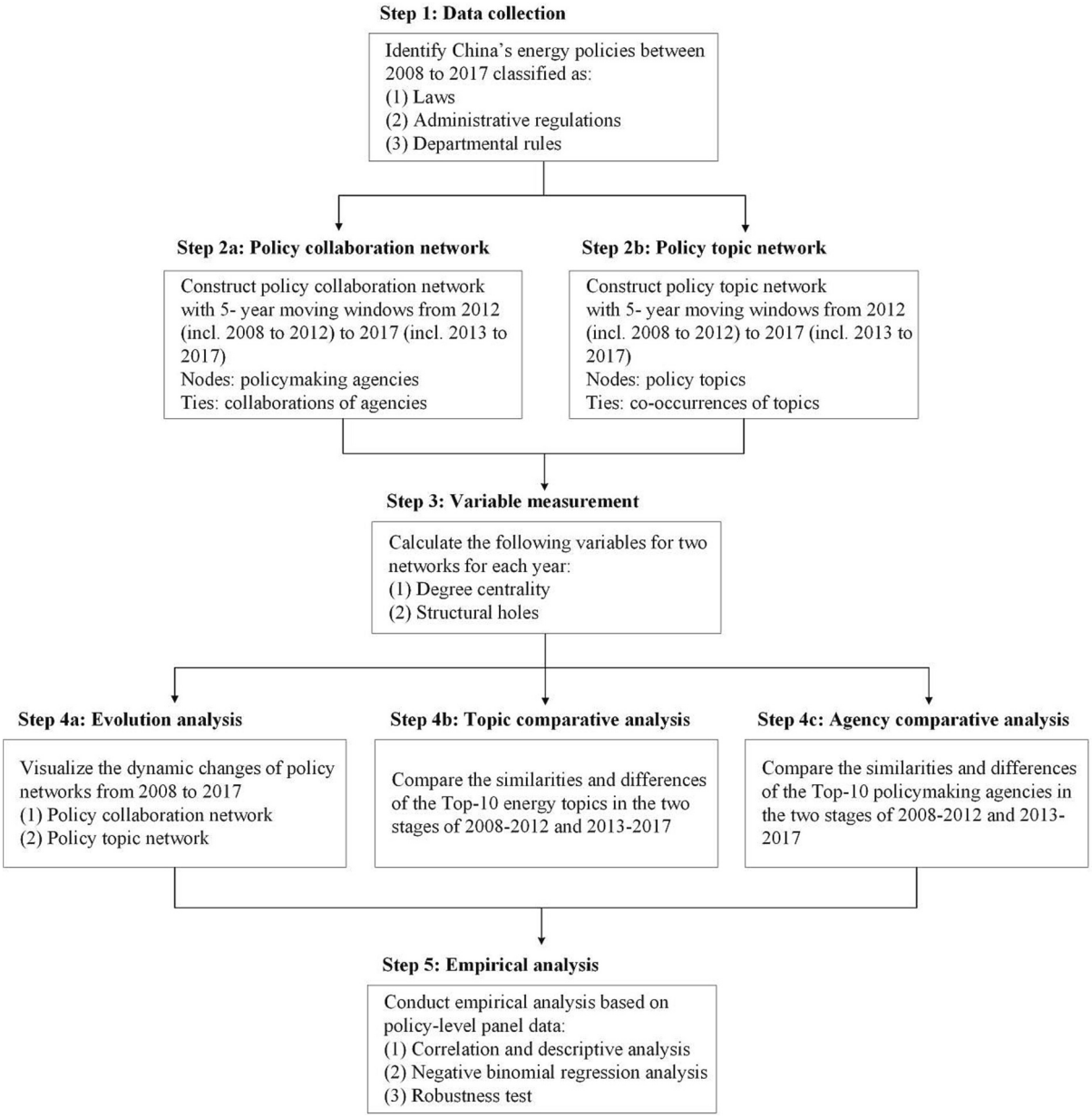


Fig. 3. Policy network analysis model.

metadata, including the following information: (1) title; (2) policymaking agencies; (3) issuing number; (4) issuing year; (5) prescription; (6) level; and (7) citation. Second, to ensure the standardization of policy documents, informal documents such as reply letters and approvals were removed, and three types of formal documents, laws, administrative regulations and departmental rules, were retained. Finally, based on the relevance principle, we manually excluded some policy documents that had the term energy in the title but were not related to energy, such as personnel transfer documents, incentive and punishment documents and numerical standards documents. Following the above screening steps, 461 energy policy documents were ultimately retained for analysis.

Policy collaboration network (CN).

First, we extracted the policymaking agency from the metadata of the policy document. Then, we unified the abbreviations of agencies and their full names. For example, we unified "General Administration for Market Regulation" and "State Administration for Market Regulation" as "State Administration for Market Regulation." Then, we used information about collaborative relationships between government agencies from the period when an energy policy was released to construct a co-occurrence matrix containing the name of the policymaking agency. Specifically, if a policy was jointly issued by two or more policymaking agencies, there was a collaborative relationship between the agencies. Finally, we constructed a policy collaboration network using policymaking agencies as nodes and collaborations as edges.

Policy topic network (TN).

High-frequency words that appear in a policy document can reflect the main message of a policy, and such high-frequency words are referred to as topics (Yang and Huang, 2020). If two topics appear together in the same policy document, there is a co-occurrence relationship between them. Before analysing such a co-occurrence relationship, it is necessary to build a corpus and perform data cleaning operations such as text splitting, merging synonyms, and removing meaningless words to eliminate interfering information and more accurately extract topics. The method of constructing a topic network was as follows.

(1) We used Lucene's SmartChineseAnalyzer (SmartCN) word divider, which is an open-source search engine based on the Chinese word segmentation system of Natural Language Processing and Information Retrieval (NLPIR), to generate discrete word distributions for each policy document.

(2) We removed stop words. SmartCN is configured with default stop words, which are mainly punctuation. Furthermore, SmartCN allows the identification of custom stop words. We added the common Chinese stop word library, which contains mainly pronouns, adverbs, prepositions, conjunctions, and adjectives, to the source file. Based on these criteria, stop words were filtered out.

(3) We used the TextRank algorithm to identify keywords as topics (Mihalcea and Tarau, 2004). According to Formula (1), the weight of each word was iteratively calculated until it converged. Then, the weights of nodes were ranked in reverse order to obtain the most important n words. If the key words formed adjacent phrases, then a key phrase was formed.

$$WS(V_i) = (1-d) + d^* \sum_{V_j \in In(V_i)} \frac{W_{ji}}{\sum_{V_k \in Out(V_j)} W_{jk}} WS(V_j) \quad (1)$$

where $WS(V_i)$ denotes the importance score of word i; $In(V_i)$ represents a collection of other words in the document that point to word i; $Out(V_j)$ indicates that word i points to a collection of other words; W_{ji} represents the weight of the edge between words i and j; and d is the damping coefficient, which is generally 0.85.

Then, we manually added some energy topics to the custom dictionary to avoid breaking down important phrases. Finally, energy policy topics were tentatively identified and organized into a "document-topics" list.

(4) We created a synonym list. The list was based on energy topics to ensure that the issue of semantic duplication did not affect the accuracy of the topics. For example, we unified "energy saving," "energy conservation" and "saving energy and reducing consumption" as "energy saving."

(5) We constructed the topic co-occurrence matrix. If two topics appeared in the same policy document at the same time, then a co-occurrence relationship existed between them. According to the cleaned "document-topics" list, we then constructed the topic co-occurrence matrix. The values in the topic co-occurrence matrix express the number of co-occurrences between topics.

(6) We constructed a topic network according to the topic co-occurrence matrix, where the nodes represent topics and the edges represent the number of co-occurrences between topics.

(7) We identified high-frequency words in each network according to the occurrence frequency of nodes, which represents the policy topics in each stage. This allowed us to compare the top 10 high-frequency policy topics over time.

3.3. Measurement

In this study, we used a longitudinal design. We built collaboration and topic networks based on moving five-year windows (i.e., 2008–2012, 2009–2013, 2010–2014…2013–2017), resulting in six snapshots of both networks. For example, we analysed policy documents published in 2012 using measures of policy network characteristics from 2008 to 2012. Our sample ultimately consisted of 461 energy policy documents, each with no less than one agency and topic. We pooled these documents at the policy-level after computing the network indicators at the level of policymaking agencies and policymakers.

3.4. Variable definitions

3.4.1. Dependent variable

The dependent variable, *policy citations*, was used to measure the degree of policy diffusion. Policy citation refers to the borrowing of a later policy from a previous policy in three ways. First, a lower-level policy cites a higher-level policy, which often means a local policy document citing a national policy document (Huang et al., 2015). Second, a new policy cites a peer policy either vertically or horizontally. Vertical citation refers to the revision and extension of a previous policy into a newer one, while horizontal citation refers to the coordination and borrowing of policies from different fields (Ba et al., 2022). Third, specific cases or journal articles cite policy documents. These policy citations are used to identify the content of the policy to capture its reference value for individual cases (Bonvecchi et al., 2016; Solberg et al., 2006). In any case, policy citation fits the definition of policy learning within the framing of policy diffusion (Wang et al., 2020). Thus, policy citations are a valuable tool for documenting a policy diffusion footprint (Ba et al., 2022).

Policy citations reflect policy adoption through citations/citation links between policies (Ba et al., 2022). Citations between policies and citations of policies by judicial cases are generally not listed as references but are included in the content of this article. Policy citations usually mark the text with quotation marks or set a citation link. On this basis, we can identify the citation relationship in the text. Journal citations of policies are normalized as references, which facilitated our statistics. In this paper, the number of policy citations is used as a measure of the degree of policy diffusion. The metadata in PKULaw contains the number and type of policy citations, covering policy and regulatory citations, judicial case citations, and journal citations. All citations are summarized in the explanatory variable used in this study, i.e., the policy citation count. For example, the policy document *Guideline of The General Office of the State Council on Accelerating the Promotion and Application of New Energy Vehicles* received 305 citations related to policies and regulations (including central regulations, local

regulations, and drafts), 18 citations related to judicial cases (including judicial cases, case reports, and law firm practice cases), and 4 citations in journals. Thus, the total number of citations for this policy was 305 + 18 + 4 = 327.

3.4.2. Independent variables

Our independent variables were the centrality and structural holes of the collaboration network and the topic network. We chose these two attributes for the following reasons: on the one hand, the above metrics are ego-level metrics that are widely used by scholars (Guan et al., 2017; Wang et al., 2014). However, global-level metrics (e.g., closeness and betweenness) may lead to suppression in the estimation of models that simultaneously analyse the joint effects of multiple central nodes, resulting in inverse coefficients (Abbasi et al., 2011).

On the other hand, centrality and structural holes have practical implications in both types of networks. In collaboration networks, the centrality of agencies represents their status and prestige (Sun and Cao, 2018). Agencies with a high degree centrality have extensive collaboration relationships and occupy a dominant position in collaboration networks. These agencies are key lead agencies for policy development. The structural holes occupied by agencies represent their connection with other agencies that do not cooperate with each other (Burt and Lazega, 1995). Structural holes indicate the control of information resources. Agencies that master many structural holes are brokers in collaboration networks. They take information and resources from one agency and transmit them to another. These brokers control the disconnected ties of the network and are integrating agencies in policymaking. Conversely, if an agency has low centrality and structural holes, it rarely establishes connections or acts as a broker. These agencies play a smaller role in policymaking and they are followers in policymaking (Yin and Huang, 2023).

In topic networks, the centrality of topics represents their combination ability. Topics with high centrality represent the popularity of policies in the domain because they often co-occur with other topics (Wang and Zhang, 2022). Identifying core topics helps policymakers understand the influential topics in the domain. Attribute characteristics of topics can provide direction for subsequent policy development. A topic's structural holes represent its degree of control over its nonredundant policy resources (Burt and Lazega, 1995). Topics that occupy many structural holes have a heterogeneous knowledge portfolio and are more likely to be combined with emerging topics to generate new ideas (Ba et al., 2021). Policymaking agencies can use this feature to explore policy innovations. The meaning of the node attributes is different between a collaboration network and a topic network. Therefore, the selection of the two attributes, centrality and structural holes, has both theoretical and practical implications. The measurement of these two sets of variables was performed in two stages: first, we calculated these network indicators using Ucinet software; then, we aggregated each network indicator at the policy-level.

3.4.3. Degree centrality of a collaboration network

This variable expresses the centrality of the position that an agency occupies, and the calculation requires two steps. First, we calculated the degree centrality of the policymaking agencies, which is the number of other agencies whose position in the network contacts a policymaking agency. Additionally, to eliminate the network's size effect, we took the normalized centrality as the value of this indicator, which is calculated as follows.

$$Nrmdegree_i = \frac{\sum_{j=1}^{g} a(i,j)}{g-1} \tag{2}$$

where $a(i, j)$ is a binary variable that indicates whether i is connected to j. If i and j are connected, $a(i, j) = 1$; if not, $a(i, j) = 0$ and g denotes the number of policymaking agencies in the policy cooperation network.

Second, we counted the policymaking agencies of each policy document since multiple policymaking agencies may be involved in a policy document. Therefore, we used the average of the centrality values of all collaborating policymaking agencies as the value of the independent variable (Wang et al., 2014).

3.4.4. Structural holes of a collaboration network

Structural holes indicate the extent to which an agency is disconnected from other agencies. We first calculated Burt's binary constraint degree C_i (Burt and Lazega, 1995). Then, based on previous experience, we subtracted the degree of constraint C_i from 2 to show the control advantage that arises from i across the structural hole (Guan et al., 2017; Wang et al., 2014).

$$Structural\ holes_i = 2 - C_i = 2 - \sum_j \left(p_{ij} + \sum_{k,k\neq i,k\neq j} p_{ik}p_{kj} \right)^2 \tag{3}$$

where i is the ego node and p_{ij} denotes the proportion of j to the total edges of i. The third node k is connected to both i and j.

Second, we aggregated the structural hole value of the agencies for each period at the policy-level and used the average structural hole value of all policymaking agencies for each policy as the independent variable.

3.4.5. Degree centrality of a topic network

The centrality of a policy topic network reflects the extent to which a particular policy topic occupies a central position. The calculation of this variable is similar to the method used to calculate centrality within a policy collaboration network.

3.4.6. Structural holes of a topic network

Suppose that the tie between policy topics is broken, and the focal policy topic happens to be located above this breakpoint. Then, we considered that the focal policy topic occupies the structural hole in the network. The calculation of this variable is similar to the calculation of structural holes in a policy collaboration network.

3.4.7. Control variables

Our study involves two distinct networks, namely, the policy collaboration network and the policy topic network. Therefore, we planned to control in two ways for variables that might be omitted and alternate factors other than network characteristics that might interfere with the measurement results. We controlled for variables related to the policy collaboration network, including the number of agencies, the level of agencies, and the number of publications. We also controlled for text-level influences, including title length and policy text length.

3.4.8. Number of agencies

Many studies have shown that the more agencies are involved in a publication, the more likely the policy is to be cited (Annalingam et al., 2014; Biscaro and Giupponi, 2014). Therefore, we controlled for possible bias caused by quantitative factors on the impact of measuring network characteristics.

3.4.9. Level of agencies

Our study concerns government agencies, and we first considered hierarchical factors in formal organizations. Numerous studies have shown that hierarchical status is an important factor influencing organizational relationships (Agneessens and Wittek, 2012; Brennecke and Rank, 2017). Therefore, we used the assignment method to distinguish between different ranks of agencies and thus controlled for the effect of institutional authority on citations. This variable takes a value of 4 if the agency takes on the leading role for the state. Since the study does not involve agencies that play the role of assisting the state, we assign a value of 3 to the next-level agency, i.e., the leading roles of ministries, a

value of 2 to the assisting roles of ministries, and a value of 1 to agencies below the assisting roles of ministries, and we used the average of the assigned values for each policymaking agency as the final value of this variable.

3.4.10. Number of publications

We used the total number of policies issued by each policymaking agency between t-5 and t-1 to control for the effect of empirical factors on citations (Carnabuci and Operti, 2013).

3.4.11. Title length

The number of words in the title correlates with the citation count (Annalingam et al., 2014; Rostami et al., 2014). Some studies have suggested that title length has a negative effect on citations (Stremersch et al., 2015).

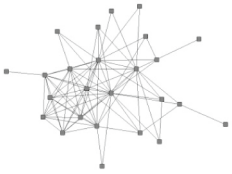
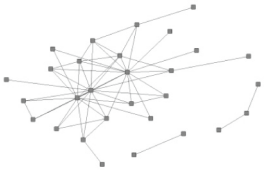
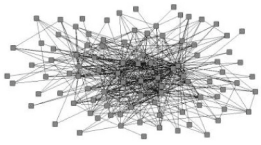
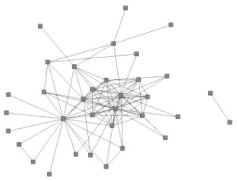
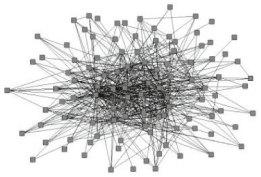
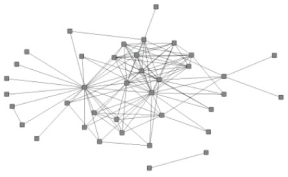
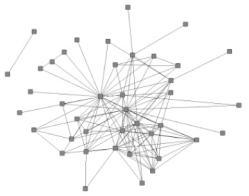
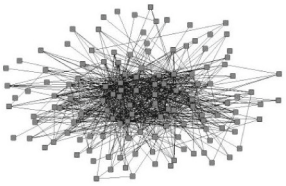
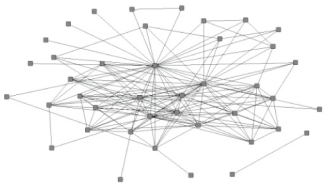
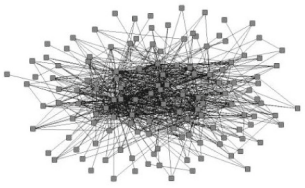
3.4.12. Policy text length

Some studies have shown that the length of the text is one of the factors that increases citations (Ayres and Vars, 2000). The longer a text is, the more citations it receives. This may be due to longer texts containing more information (Leimu and Koricheva, 2005). Therefore, we speculated that policy text length might have the same effect on policy citations.

3.5. Model specification

Since the number of policy citations used for the dependent variable in this paper is a nonnegative integer and the variance (SD = 33.978) is

Table 4
Dynamic change in policy collaboration networks and policy topic networks.

	Policy collaboration network	Policy topic network
2008-2012		
2009-2013		
2010-2014		
2011-2015		
2012-2016		
2013-2017		

larger than the mean (Mean = 10.449), it has the characteristics of dispersion. Therefore, we chose the negative binomial model to test the empirical results.

The specific model is as follows.

$$Policy\ citations_{i,t} = exp\Big(\sum \beta_0 + \beta_1 Controls + \beta_2 CNNrmDegree_{i,t-1} + \beta_3 CNStructural\ holes_{i,t-1} + \beta_4 CNStructural\ holes^2{}_{i,t-1} + \beta_5 TNNrmDegree_{i,t-1} + \beta_6 TNStructural\ holes_{i,t-1} + \mu_{i,t-1}\Big) \quad (4)$$

where we analyse year t using measurements from t-5 to year t-1. *Policy citations*$_{i,t}$ represents the number of citations of policy document *i* from year *t* to the observation cut-off year (2020). *Controls* denotes control variables. *CNNrmdegree*$_{i,t-1}$ indicates the centrality indicator of the policy collaboration network. *CNStructural holes*$_{i,t-1}$ denotes the structural holes of the policy collaboration network. *TNNrmdegree*$_{i,t-1}$ represents the centrality indicator of the policy topic network. *TNStructural holes*$_{i,t-1}$ denotes the structural holes of the policy topic network, with *i* denoting the policy document, *t* denoting the year, $\beta 1 \sim \beta 5$ being the regression coefficients, and $\mu_{i,t-1}$ being the random error term.

4. Results

4.1. Description analysis of policy networks

4.1.1. Evolution of the policy collaboration network and policy topic network

We visualized the evolutionary trend maps of the energy policy collaboration network and the topic networks through two separate sets of figures. We created six snapshots of each network based on a 5-year moving window for the 10-year period spanning 2008–2017, as shown in Table 4.

The number of nodes in both networks gradually increased over time, indicating that more agencies became involved in energy policymaking activities and the policy topics became richer. In terms of linkages, there were more connections between agencies, and the scale of the networks expanded. Similarly, there was a substantial increase in linkages between topics, creating greater cohesion within the topic network.

4.1.2. Comparison of top 10 energy topics in different periods

We calculated the frequency, degree centrality and structural holes of energy topics during the periods 2008–2012 to 2013–2017 to identify the top 10 energy topics over these two periods (Table 5). By comparing the evolution of the energy theme, we obtained the following observations.

(1) "Energy," "Electricity" and "Policies and regulations" are the enduring core topics of China's energy field. "Coal" ranked seventh and sixth in the two networks. This indicates that although the national dominance of coal will not change for a long time, its nonrenewable nature means that its development potential is not as good as that of renewable energy such as electricity. The Chinese government attaches great importance to the use of "Policies and regulations" to promote the development of the power industry and works to actively reform the power system. For example, the issuance of *Several Opinions of the CPC Central Committee and The State Council on Further Deepening the Reform of the Electric Power System (No. 9, 2015)* launched a new round of electricity system reform. Notably, the term "Coal" has relatively low centrality but occupies a high structural hole. This reflects the dominance of coal in China's energy mix.

(2) The status of the terms "Security" and "Supervision" increased. Changes in topics from 2013 to 2017 revealed that security and supervision jumped to second, behind only energy and electricity. In today's world, designing a solution for energy security is the common responsibility of the international community. Ensuring energy security has become a top priority for energy development in China, which is the world's largest energy consumer. At the same time, the Chinese government attaches great importance to the supervision of the energy market and industry to maintain a fair and just energy market order.

(3) Emerging topics were "Clean energy" and "Production." This shows that the Chinese government has made great efforts to develop clean energy and promote the transformation of the energy structure from high carbon to low carbon and from fossil energy to clean energy. On the energy supply side, the aim is to build a diversified clean energy supply system. With the continuous improvement of energy supply capacity and level, China has become the largest energy producer in the world.

4.1.3. Comparison of top 10 policymaking agencies in different periods

We ranked government agencies according to the number of published documents and calculated their corresponding levels of degree centrality (Table 6). When we compared the status of core policymaking agencies over the two periods, 2008–2012 and 2013–2017, we found little overall change in the core agencies. The "National Energy Administration," the "Ministry of Finance" and the "National Development and Reform Commission" were the three core agencies issuing energy policies. The number of joint papers issued by these three agencies, as well as other agencies, also ranked among the top three, indicating that they dominated the development of China's energy industry. The big change was the "Ministry of Environmental Protection." The agency did not appear in the rankings in the former period but replaced the "State Electricity Regulatory Commission" as the core agency in the latter period. This change shows that China's energy development followed a kind of green development, paying attention to both the unity of energy development and environmental protection.

4.2. Estimation results

Table 7 illustrates the correlations among the variables. As all variance inflation factor (VIF) values are much <5, there is no significant

Table 5
Top 10 energy topics (2008–2012, 2013–2017).

2008–2012					2013–2017				
Rank	Topics	Frequency	NrmDegree	Structural holes	Rank	Topics	Frequency	NrmDegree	Structural holes
1	Energy	105	0.107	1.893	1	Energy	259	0.057	1.943
2	Fiscal policy	86	0.056	1.944	2	Electricity	232	0.034	1.966
3	Electricity	83	0.050	1.950	3	Security	174	0.018	1.982
4	Technology	51	0.054	1.946	4	Supervision	116	0.022	1.978
5	Quality	46	0.036	1.964	5	Policies and regulations	104	0.020	1.980
6	Policies and regulations	40	0.039	1.961	6	Coal	89	0.011	1.989
7	Coal	38	0.015	1.985	7	Technology	88	0.022	1.978
8	Clean energy	36	0.035	1.965	8	Production	82	0.015	1.985
9	Energy saving	36	0.041	1.959	9	Quality	65	0.011	1.989
10	Supervision	30	0.030	1.970	10	Clean energy	63	0.014	1.824

Table 6
Top 10 policymaking agencies (2008–2012, 2013–2017).

2008–2012				2013–2017			
Rank	Agencies	Frequency	NrmDegree	Rank	Agencies	Frequency	NrmDegree
1	National Energy Administration	76	0.125	1	National Energy Administration	318	0.063
2	Ministry of Finance	43	0.118	2	National Development and Reform Commission	67	0.059
3	National Development and Reform Commission	41	0.144	3	Ministry of Finance	28	0.032
4	State Electricity Regulatory Commission	16	0.034	4	Ministry of Water Resources	23	0.015
5	Ministry of Housing and Urban-Rural Development	12	0.034	5	Ministry of Industry and Information Technology	20	0.022
6	State Council	12	0.016	6	Ministry of Environmental Protection	18	0.026
7	Ministry of Water Resources	11	0.017	7	State Council	18	0.008
8	Ministry of Industry and Information Technology	9	0.034	8	Ministry of Housing and Urban-Rural Development	14	0.021
9	State Administration of Coal Mine Safety	9	0.036	9	State Administration of Coal Mine Safety	11	0.010
10	General Administration of Quality Supervision, Inspection and Quarantine of the People's Republic of China	8	0.024	10	General Administration of Quality Supervision, Inspection and Quarantine of the People's Republic of China	9	0.011

multicollinearity (Belsley et al., 1980). The correlation between the variables in the collaboration network and those in the topic network is low. The correlation between the normalized degree centrality in these two networks is 0.226. The correlation between the structural holes in these two networks is 0.156. This confirms that the collaboration network is loosely coupled with the topic network.

Table 8 shows the results of a negative binomial regression of the dependent variable, *policy citations*, on the independent variables. Model 1 contains only four control variables. Then, explanatory variables were successively added in Models 2–5. All variables are included in the full model, which is Model 6.

As predicted by Hypothesis 1, the centrality of a policymaking agency in a collaboration network is positively related to its citation count. The more central a government agency is in the collaboration network, the more influential the policy documents it develops and the more citations it receives. Model 2 in Table 8 provides support for this hypothesis at the significance level of $p < 0.01$ (b = 15.848, p < 0.01). When all other variables remain constant, a one-standard-deviation increase in TN degree centrality is associated with a change in the number of policy citations of 1.797 (=exp.[0.037*15.848]), which is approximately 5 % of the standard deviation of the number of policy citations.

Model 3 shows a significant inverted U-shaped effect of structural holes in the collaboration network of policymaking agencies on policy citations. The initial increase in the variable tends to improve the quality of decisions and increase the number of citations. However, further increases may reduce the number of policy citations by falling into a "capacity trap." Previous studies have proposed a three-step procedure to test this relationship (Haans et al., 2016; Lind and Mehlum, 2010). First, the coefficient of the square term of the CN structural holes needs to be significant and negative. Model 3 in Table 8 shows that the square term of the CN structural holes is negative and significantly correlated with policy citations (b = −5.064, $p < 0.05$). Second, the slopes at both ends of the data range must move in different directions. The range of CN structural holes is between 0 and 1.696. The slope was significantly positive at value 0 (β = 17.012, p < 0.01) and negative at value 1.696 (β = −4.937, p < 0.05). Third, the turning point needs to be within the data range. The calculated turning point is 1.152, which is on the right side of the range of values for this variable. This three-step procedure confirms that there is an inverted U-shaped relationship between the CN structural hole and policy citations. An assessment of the practical effect indicates that for the agency with a CN structural hole of 1.535 (i.e., average), policy citations decrease by 2.854 for each unit increase in the CN structural hole. For the agency with CN structural holes one standard deviation (SD) below the average (1.238), policy citations increase by 0.99 for each unit increase in CN structural holes. For the agency with CN structural holes one SD above the average (1.832), policy citations decrease by 6.698 for each unit increase in CN structural holes. These findings support Hypothesis 2.

Hypothesis 3 predicts that the centrality of policy topics in a topic network is positively correlated with the number of policy citations. Model 4 supported this hypothesis (b = 33.828, $p < 0.01$). The more central the policy topics are, the more often their policies are cited. An assessment of the practical effect reveals that a one-standard-deviation increase in TN degree centrality is associated with a change in the number of policy citations of 1.403 (=exp.[0.01*33.828]), which is approximately 4 % of the standard deviation of the number of policy citations. Hypothesis 4 predicts a positive correlation between the structural holes regarding policy topics in the topic network and the number of policy citations. Policy topics that occupy rich structural holes in their policy topic networks receive more citations. The regression results presented in Model 5 show nonsignificant coefficients for this variable. Therefore, Hypothesis 4 is not supported.

observations = 461.

4.3. Robustness test

4.3.1. Changing the sample size

Since there are some extreme values in the dependent variable policy citation count, the regression results of the core explanatory variables remain significant after the bilateral tailing of the data at the 5 % quantile to verify the robustness of the previous section. The conclusions of the four hypotheses remain consistent with the previous findings after bilateral tailoring at the 5 % quantile for the number of policy citations.

4.3.2. Changing the independent variable metric

Since there is more than one measure of structural holes, we use efficiency to replace structural holes in the previous section to test the robustness of the results. The effect of the centrality of the collaboration network and the centrality of the topic network on the number of policy citations is significantly positive. The structural holes in the collaboration network have an inverted U-shaped effect on the number of policy citations. However, the structural holes in the topic network have no significant effect on this value. Overall, this is consistent with previous findings and further validates the robustness of the results.

4.3.3. Controlling event interference

Since the period we selected for the study included a national leadership change event that occurred in 2013, this event constituted an essential factor influencing policy orientation. Therefore, to further verify the robustness of the model findings and to mitigate the effect of data structure on the regression results, we defined dummy2013 as a dummy variable. We set any time from 2013 and before to 0 and any time after 2013 to 0. Then, we added dummy variables to the model for regression analysis. The results remain consistent with those of the

Table 7
Correlations and descriptive statistics.

Variables	Min	Max	Mean	S.D	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Policy citations	0	501	10.45	33.98	1.000								
(2) CNNrmDegree	0	0.129	0.071	0.037	−0.091*	1.000							
(3) CNStructuralholes	0	1.696	1.535	0.297	0.031	0.517***	1.000						
(4) TNNrmDegree	0.002	0.107	0.021	0.01	0.060	0.226***	0.001	1.000					
(5) TNStructuralholes	0.924	1.871	1.747	0.114	0.066	0.023	0.156***	−0.101**	1.000				
(6) Number of agencies	0	10	1.508	1.147	0.115**	−0.186***	−0.002	−0.131***	0.043	1.000			
(7) Title length	14	128	34.84	10.44	−0.061	−0.316***	−0.182***	−0.053	−0.025	0.451***	1.000		
(8) Level of agencies	1	4	2.288	0.526	0.277***	−0.441***	−0.024	0.009	0.007	0.303***	0.136***	1.000	
(9) Policy text length	3	25,053	2498	2647	0.101**	−0.025	−0.002	0.064	0.130***	0.062	−0.031	0.159***	1.000
(10) Number of publications	0	318	146.8	107.5	−0.166***	0.322***	0.542***	−0.068	0.078*	−0.319***	−0.247***	−0.511***	−0.132***

Note: Observations = 461.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

previous section.

5. Discussion

Policy research should focus on the level of not only the policymaker but also the policy text. Therefore, we explored the antecedents of policy diffusion from the perspective of policy collaboration networks and policy topic networks. To test our hypotheses, we used normative energy policy documents found on official government websites. Through word segmentation, deduplication, cleaning, and other text-mining techniques, we extracted the agencies and topics from each document. Then, we constructed a policy collaboration network using policymaking agencies as nodes and a policy topic network using topics as nodes. We then measured the centrality and structural holes in each of the two networks separately and pooled the measurement results at the policy-level. Finally, we applied a negative binomial regression model to empirically test the research hypotheses, exploring the form of policy collaboration networks and policy topic networks on policy diffusion. Our analysis yields the following findings.

First, our analysis of the evolution of the energy policy network reveals that China's energy issues are characterized by strong policies, many involved agencies, and much collaboration. Therefore, the Chinese government should strengthen macrocontrol over energy work and conduct long-term, focused and overarching policymaking, supervision and management of energy issues. In addition, our comparative analysis of the evolution of policy topics indicates that China's energy field has undergone profound changes. The Chinese government has vigorously promoted the building of a clean, low-carbon, safe and efficient energy system. In the future, China should further adjust its energy structure and develop clean energy to cope with the current situation of energy shortages and complex changes in the international energy market. Production departments should develop technology to improve energy efficiency, and the government should promote the energy revolution to ensure energy security.

Second, our empirical analysis shows that the centrality of agencies in policy collaboration networks positively affects policy diffusion, and the structural holes associated with agencies in those networks have an inverted U-shaped effect on the degree of policy diffusion. These results offer some cautionary notes for China's energy policymakers. On the one hand, the status of the policymaking agencies is not the only determinant of the degree of policy diffusion; the heterogeneity of the resources at the disposal of the sector is also an important factor. Our empirical study demonstrates that both the centrality and the structural holes of policymaking agencies have significant effects on policy diffusion, suggesting that both the status and the heterogeneity represented by the two elements are of great significance. On the other hand, our findings show that the centrality of policymaking agencies is positively associated with the degree of policy diffusion. This suggests that government agencies need to keep abreast of current events, grasp policy topic trends, enrich their knowledge base, expand the breadth of their policy knowledge, and build a rich resource base of policy topics to ensure policy innovation. However, the inverted U-shaped characteristic of the collaboration network warns government agencies to avoid falling into the "capacity trap," it is important to consolidate and improve existing policy elements while expanding the policy elements reserve and keeping that reserve within a moderate range.

Third, we empirically examine how policy diffusion is related to policy networks at the topic level. We find that the centrality of policy topics is positively related to the degree of policy diffusion, while structural holes of topics are not significantly related to the degree of policy diffusion. This indicates that enriching the coverage of policy topics is conducive to expanding policy diffusion. When formulating policies, government agencies should focus on the connection between different policy topics and improve the diversity of policy topics. They should not be limited to only current topics. Moreover, these agencies should consider the problem from the perspective of linkage, integrate

Table 8
Regression results for policy citations.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Number of agencies	0.784***	0.722***	0.611***	0.742***	0.780***	0.497***
	(5.253)	(4.892)	(3.682)	(5.162)	(5.266)	(3.374)
Title length	−0.059***	−0.054***	−0.053***	−0.056***	−0.059***	−0.051***
	(−6.064)	(−5.502)	(−5.879)	(−5.992)	(−6.087)	(−5.890)
Level of agencies	0.965***	1.431***	0.805***	0.926***	0.951***	1.288***
	(4.283)	(6.367)	(3.178)	(4.039)	(4.235)	(3.975)
Policy text length	0.000**	0.000*	0.000**	0.000**	0.000*	0.000**
	(1.965)	(1.881)	(2.012)	(2.031)	(1.891)	(2.093)
Number of publications	0.001	−0.002	−0.001	−0.001	0.001	−0.001
	(0.442)	(−1.124)	(−0.509)	(−0.349)	(0.459)	(−0.547)
CNNrmDegree		15.848***				10.954*
		(3.647)				(1.779)
CNStructuralholes			14.770***			17.012***
			(2.795)			(3.010)
CNStructuralholes*CNStructuralholes			−5.064**			−6.471***
			(−2.374)			(−2.812)
TNNrmDegree				33.828***		29.582***
				(3.191)		(2.786)
TNStructuralholes					0.638	0.089
					(0.770)	(0.112)
Constant	−0.120	−2.498***	−10.006***	−1.141	−1.190	−13.323***
	(−0.174)	(−2.999)	(−3.062)	(−1.532)	(−0.768)	(−3.304)
Log likelihood	−1318.5834	−1309.4309	−1301.5962	−1312.6541	−1317.9956	−1293.7591
Wald chi^2	110.14***	140.22***	122.15***	125.33***	112.45***	137.80***

Note: t-statistics in parentheses.
*** $p < 0.01$, ** $p < 0.05$, *$p < 0.1$.

the entire process, address multidisciplinary and multilevel refracted problems, and formulate comprehensive and scientific decisions.

6. Conclusion and outlook

Our study makes several contributions. First, it advances the related research on the consequences of policy networks. We promote the propositions of Holden and Lin (2012) and Füglister (2012) and build a text-based policy network research framework that provides a new perspective for analysing the antecedents of policy diffusion. Second, we contribute to the understanding of the multiple natures of policy networks. Following the outlook of Ba et al. (2022), we construct policy collaboration networks and policy topic networks with more granular and precise policy elements (policymaking agencies and policy topics) to explore the different effects of their structural features from a comparative perspective. Finally, we provide policy advice to the rapidly developing energy field in China by observing the evolution of core energy topics and policymaking agencies. The results also provide a reference for energy development in other countries.

The potential applications of this approach are as follows. First, this article provides a general approach to building collaboration networks and topic networks. Joint publications exist not only in the energy field but also science (Guan et al., 2016), technology (Sokolov et al., 2019), agriculture (de Arroyabe et al., 2021), and other research fields. This method is universal when there is a collaborative relationship within the research field, and the collaboration content forms an observable document. In addition, the research framework of this paper can be used to explore policy networks in other countries. Joint policies/agreements are considered to be effective measures to maintain collaborative relationships (Steen-Adams and Charnley, 2023). Policy collaboration occurs not only in China but also in other countries (such as the United States (Steen-Adams and Charnley, 2023), the United Kingdom (Georgalakis, 2020), Germany, and Finland (Wagner et al., 2021)). Such collaboration takes place not only within a country but also internationally, such as among BRICS countries (Sokolov et al., 2019) and members of the European Union (Grande and Peschke, 1999). By constructing collaboration networks, researchers can better understand the evolution of organizational structures and identify the core agencies in different countries. In addition, other countries can extract topics from policies/agreements to build policy topic networks, which may help researchers grasp the policy direction of the country and interpret the policy results from a new network perspective. Future work can explore comparative investigations of collaborative networks and topic networks across different fields and countries.

Our study also has some limitations. First, our analysis is based exclusively on data from the energy field. Future research could extend our findings to other fields. Second, we use the number of policy citations to measure policy diffusion, which to some extent ignores the differences among citation targets. Future studies could distinguish among central regulation citations, local regulation citations, judicial case citations, and so on. Bridging these limitations in future studies will facilitate the exploration of the relationship between policy networks and policy outcomes. We believe that this research provides valuable insights despite these limitations.

Author statement

On behalf of all other authors, we declare that we have no conflicts of interest. We certify that the submission is original work and is not under review at any other publication.

Data availability

Data will be made available on request.

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